

Artificial Neural Networks Assignment

Customer Churn Prediction in Telecommunications

1. Introduction

The objective of this analysis is to examine customer behaviour using a telecommunications dataset and build a neural network model to predict whether a customer is likely to churn. In addition to predictive modelling, the goal is to interpret the patterns in the data and translate them into actionable business insights.

The dataset contains information about individual customers such as demographic characteristics, services subscribed, billing details, and contract information. Each record represents one customer and the target variable indicates whether the customer has churned.

2. Data Understanding and Preparation

The dataset contains **7043 customer records and 21 variables** describing customer characteristics and service usage. These variables include demographic information (such as gender and senior citizen status), service subscriptions (internet service, streaming services, device protection), contract type, billing details, and payment method.

The prediction task is a **binary classification problem**, where the model predicts whether a customer will churn (1) or remain with the service (0).

Data Cleaning

During preprocessing, the following steps were performed:

1. **Handling missing values**

The TotalCharges column contained missing entries represented as blank strings. These values were converted into numeric format and rows with missing values were removed.

2. **Data type conversion**

The TotalCharges variable was converted from string format to a numerical variable so that it could be used in the model.

3. **Encoding categorical variables**

Many features such as contract type and internet service are categorical. These variables were converted into numerical representations using one-hot encoding so they could be used by the neural network.

4. **Feature scaling**

Numerical features such as tenure, monthly charges, and total charges were scaled using standardization. Scaling ensures that all features contribute equally during model training.

5. **Feature selection**

The customerID column was removed because it does not provide predictive information and simply acts as a unique identifier.

After preprocessing, the final dataset used for modelling contained **7032 records and 30 input features**.

3. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand behavioural patterns associated with customer churn.

Insight 1: Contract type strongly influences churn

Customers with **month-to-month contracts show significantly higher churn rates** compared to customers with one-year or two-year contracts. Longer contracts appear to increase customer commitment and reduce the likelihood of leaving the service.

This insight suggests that contract structure plays an important role in customer retention.

Insight 2: Customers with low tenure churn more frequently

The analysis shows that customers who have been with the company for a shorter period are more likely to churn. Many churn cases occur during the early months of the customer lifecycle.

This suggests that the **initial customer experience is critical**. Poor onboarding experiences or unmet expectations early in the relationship may lead customers to leave.

Insight 3: Higher monthly charges are associated with higher churn

Customers with higher monthly charges appear more likely to churn compared to customers with lower charges.

This pattern suggests that customers paying higher fees may have higher expectations regarding service quality and perceived value. If these expectations are not met, they may be more inclined to switch providers.

4. Neural Network Modelling

A **feedforward artificial neural network** was used to predict customer churn.

The model architecture consisted of:

- Input layer with all selected features
- First hidden layer with **16 neurons** using the ReLU activation function
- Second hidden layer with **8 neurons** using the ReLU activation function
- Output layer with **1 neuron** using the Sigmoid activation function

The sigmoid function is appropriate for binary classification because it produces probability values between 0 and 1.

Loss Function

The model used **Binary Cross-Entropy** as the loss function because the problem involves predicting two classes (churn vs non-churn).

Optimizer

The **Adam optimizer** was used because it efficiently adjusts learning rates during training and performs well for many neural network problems.

The dataset was split into training and testing sets to evaluate the model's ability to generalize to unseen data.

5. Evaluation and Interpretation

The trained neural network achieved an overall **accuracy of approximately 79%** on the test dataset.

The confusion matrix results show that the model performs well in identifying customers who are likely to stay. However, it struggles more when predicting customers who will churn.

The classification metrics indicate:

- Precision for churn prediction ≈ 0.64
- Recall for churn prediction ≈ 0.49
- F1-score ≈ 0.55

The relatively lower recall indicates that the model fails to detect some customers who eventually churn. These cases represent **false negatives**, where the company misses the opportunity to intervene and retain the customer.

Despite this limitation, the model successfully identifies a substantial number of churn cases and provides useful predictive insights for business decision-making.

6. From Insight to Decision

Based on the analysis, two actionable strategies can be proposed.

Decision 1: Encourage longer-term contracts

Since customers on month-to-month contracts exhibit significantly higher churn rates, the company could introduce incentives that encourage customers to move to longer contracts. These incentives may include discounted rates, bundled services, or loyalty benefits.

This strategy aims to increase customer commitment and reduce the likelihood of churn.

Risks and assumptions

This approach assumes that customers are willing to commit to longer contracts if provided with incentives. However, if competitors offer more flexible plans, customers may still prefer short-term contracts.

Decision 2: Improve value perception for high-charge customers

Customers paying higher monthly charges appear more likely to churn. This indicates that perceived value plays an important role in retention.

The company could implement targeted programs for customers with higher monthly bills, such as personalized offers, improved customer support, or additional service benefits.

Risks and assumptions

This strategy assumes that churn among high-paying customers is related to value perception rather than external factors such as competitor pricing or relocation.